

CLOCK IN - A SOLANA MOBILE HACKATHON

LAST MAN SKATING

Solana Mobile Edition

Proof of Play: earn by skating, not gambling.

New: Proof of Meet (co-signed NFTs), a Seeker world, card gifting, a wallet space and SKR holder detection.

KRISTEN STUDIOS GAMES | Track: Clock In Hackathon | Updated 4 Oct 2026 (v3)

A live, organically-growing mobile game

An isometric skating battle royale built in HTML5 canvas, packaged natively for Android via Capacitor - already published and growing before this hackathon began.

3000+

organic installs

on Google Play

9

playable worlds

lake, space, volcano, Japan, deep sea and more

57

collectible skins

5 rarities, unlocked by play or offered in person

PvP

online duel & FFA

AI matches are tracked separately and do not feed the pool

Porting a live game onto Solana Mobile

1 Native MWA plugin

Capacitor + Kotlin plugin wired directly to Mobile Wallet Adapter: no deep links, any MWA wallet, Seed Vault detection, automatic re-authorization when a wallet session expires.

2 Proof of Play program

Anchor program: weekly pool, validated scores from a trusted authority, finalization and claim. 75% by score, 25% shared equally, double-claim impossible.

3 Automated settlement

Cloudflare Worker reads weekly Firebase stats, submits scores on-chain, finalizes the week, then opens and funds the next pool - every Monday, with a protected manual override.

4 Proof of Meet

Two players physically meet (Nearby), both wallets sign the same transaction, and two "Sanctuaire Seeker" NFTs are minted atomically. Plus Seeker world, card gifting and a wallet space.

Also new: read-only SKR holder detection on mainnet (see slide 08).

HOW IT WORKS

From match to claimed LMS, every week

1 PLAY

Online Duel, FFA or RADAR: real-time PvP against human opponents.

2 SCORE

Weekly wins are a delta against last week's snapshot, capped per player (21 duel / 14 FFA wins).

3 SETTLE

The Worker submits scores on-chain and finalizes the weekly pool.

4 CLAIM

The player signs with any MWA wallet. Up to 9 weeks are caught up in one go; success is shown only after on-chain confirmation.

Scores come from player-declared counters in Firebase, bounded by the weekly cap; verified opponent pairs are on the roadmap.

Proof of Play, not chance

No wagering. No randomness on the payout amount. The weekly pool is distributed purely by real competitive performance.

75%

SPLIT BY SCORE

Proportional to each player's weekly Duel + FFA wins.

25%

SHARED EQUALLY

Among every active player above a minimum score.

Meet in person. Sign together. Mint together.

- Google Nearby Connections (BLE + Wi-Fi) puts two real players in contact.
- Both wallets must sign the SAME transaction - the double signature is the on-chain proof of the meeting.
- Two Metaplex Core NFTs are minted atomically: all or nothing.
- Premium tier only if both players have Seed Vault (Seeker); Classic otherwise. On-chain caps: 1,000 Premium / 10,000 Classic.
- One mint per wallet, enforced by a PDA record. A durable nonce removes blockhash-expiry failures in a two-human flow.

NFT unlocks the Seeker skin

Ownership is verified on-chain through Helius DAS (metadata URI match), not by a Firebase flag a player could write. Tested end to end on two physical devices.

Seeker world, card gifting, wallet space

Solana Seeker world

RADAR duels replay the frozen lake with a dedicated Solana Seeker look and music. A visual skin only - no gameplay change - armed on both devices through the invite.

Card gifting

57 skin cards in 5 rarities. Give one to a player standing next to you over Nearby: no server, no NFT. The receiver accepts or declines. Daily cap by receiver: 3 cards on Seeker, 1 on standard Android.

Wallet space: MES NFT

In Settings: LMS balance, a grid of the NFTs you own, a music filter, and a detail view with attributes and an audio player. Read-only (no signature), metadata sanitized.

Read-only SKR holder detection

When a wallet is connected, the game reads its SKR balance on **mainnet** (read-only `getTokenAccountsByOwner`, via Helius mainnet with public RPCs as fallback). **No signature, no transaction**: the game itself still runs on devnet.

OFFICIAL SKR MINT

SKRbvo6Gf7GondiT3BbTfuRDPqLWei4j2Qy2NPGZhW3

solanamobile.com/skr

"SKR Holder" skin

Unlocked when the wallet holds at least 1 SKR.
Appears in the Seeker skins section of the shop.
Same dash power as the Seeker skins (a fan of 13 magic-fist arcs, red look). Many other skins also have a dash power: nothing SKR-specific.

"SKR : n" badge

Shown next to the LMS balance in the "MY NFTs" wallet area, so holders see their SKR balance next to their game assets.

Client-side, read-only

The unlock is checked on the client, like the other Seeker skins (known, accepted limitation). Dash power has limited impact on play; the weekly pool depends only on the server-side score: no stake, no draw.

NOT IMPLEMENTED YET (MAINNET ROADMAP)

Weekly Proof of Play pool funded in SKR, and SKR-priced skins. Not testable on devnet by judges (SKR only exists on mainnet): see the demo video.

The stack, end to end

ON-CHAIN

Anchor program on devnet, 7 instructions: init pool, fund, submit, finalize, claim, init mint counter, co-signed NFT mint. Hand-encoded Metaplex Core CPI, u128 payout math, PDA anti-double-claim and anti-double-mint. 8/8 unit tests.

BACKEND

Cloudflare Worker + Firebase RTDB + KV snapshots. Weekly cron at 00:01 UTC: settle, open and fund the next pool. Protected manual endpoints. Confirmation timeouts are verified on-chain before being treated as failures.

MOBILE

Capacitor + native Kotlin MWA plugin, Seed Vault / Seeker compatible, Nearby Connections, Helius DAS for NFT ownership, read-only SKR balance check on mainnet. HTML5 canvas game core, Android TV remote support.

VALIDATED, NOT JUST DEMOED

Tested end-to-end on real devices

Every critical instruction was run on physical Android devices (Phantom and the Seeker's Seed Vault) on Solana devnet, not only in Playground unit tests.

- Wallet connect and authorize (Mobile Wallet Adapter)

- fund_pool: weekly pool funded, now automatic

- Double claim rejected (AlreadyClaimed)

- Co-signed NFT mint on two devices: tablet + Seeker (20 Sep)

- submit_score confirmed on Solana Explorer

- claim_scratch: tokens received in the player wallet

- Multi-week claim with on-chain confirmation (30 Sep)

- Weekly cron: settlement + next pool auto-opened

Documented honestly, tackled next

Anti-collusion scoring

Score matches against verified opponent pairs instead of self-declared totals.

Scores by wallet

Group scores by wallet, not by anonymous uid, so a reinstall never splits a player's week.

On-chain claim deadline

Add an expiry and a redistribution path for unclaimed pools.

Mainnet + SKR

Deploy the real LMS mint, a dedicated RPC endpoint and a local web3.js bundle. Weekly pool funded in SKR and SKR-priced skins.

Fair mint fees

Split the mint cost 50/50 between the two players.

Two store builds

Separate Play Store and dApp Store builds.

Built solo. Shipped for real.

Moleouf - Kristen Studios Games | github.com/moleouf/last-man-skating-solana | Last Man Skating, live on Google Play